goals. It has been designed to reduce power consumption.

These two radio types are for two distinct cases. Bluetooth Mesh is not a radio but a networking stack. It enables thousands of devices to communicate using Bluetooth Low Energy as an underlying Bluetooth technology.

Mesh networking

A mesh network is like a spider's web where any device can talk to any other device, as opposed to Wi-Fi, which is a hub router with some spokes, where every communication takes place by going through the router.

Devices that are members of the mesh network (nodes) can communicate with other devices, and these do not have to be in direct radio range. This is accomplished by sending messages across the network by hopping from device to device until they reach their final destination. So, the problem of range is solved through a mesh topology and multi-hop delivery.

With this idea, an entire building can be encompassed with a mesh network where messages reach their destination, travelling at the speed of sound. Latency is very low with Bluetooth Mesh. Even a collection of buildings like business campuses and universities can all have an entire mesh network.

Bluetooth Low Energy has a range of hundreds of metres in clear sight, but is smaller in a building. Meanwhile, Bluetooth Mesh allows messages to travel up to 127 hops.

Multi-path is more unique to Bluetooth, and the way mesh networking is done. One of the primary goals while designing the mesh network is its reliability. It is hard to create a wireless communication technology that is sufficiently reliable for large-scale commercial lighting systems. Often, there are multi-cast operations and a large group of devices are addressed in one go. Technologies like IP do not work well because of the way acknowledgments work in their networking stack. Thus, the network gets flooded very quickly.

In multi-path delivery, any message sent by a device—for example, a message sent by a light switch—will travel through multiple paths through the mesh network to arrive at its destination. Even if one of those paths is blocked by a physical barrier or a device is broken, the function will not be affected. A copy of the message will get there. So, a very high level of availability and reliability can be accomplished through such mechanisms.

Bluetooth Mesh

Bluetooth Mesh is inherently a secure approach to mesh networking. It is not good for things like ad-hoc networking because devices that are not members of a network cannot communicate in it. Consequently, there is a membership process called provisioning. To use Bluetooth, you need to obviously pair devices. But in Bluetooth Mesh, you do not pair devices because it is not about relationships between only two devices; it is many to many. So, standard pairing will not scale. It will take forever if pairing has to be done separately for each light or thermostat in the building, and this is why the new provisioning process was created.

Provisioning can be carried out using a smartphone or tablet application. For provisioning, an unprovisioned device, which has just been taken out of its box, is equipped with various security keys that make it a member of the network and allow it to interact with other parts of the network securely.

During provisioning or immediately after, some configuration can be done using the application provided by the manufacturer. And while doing so, it is important to specify which application it is a part of, because the network hosts different applications. So, in a building context, heating, lighting and physical security can be some of the applications. And particular types of devices will participate in one or more of those applications.

Different ways of addressing messages

Devices have what we call states, which really are data items that indicate or control different states or conditions that the devices can be in. A very simple idea would be 'on' or 'off'. That will be represented by a state that has a 1 or 0 in it. And state data items are all in specifications where they are all defined. So, values that they can take, what those values mean, how these can be changed and what should happen are all defined, because this is a standard design that allows interoperability between devices from different manufacturers.

Interoperability usually means integration and not devices that work together just because they work based on the same standards. Instead, devices can be made to be interoperable by writing custom code. However, this definition is not applicable in the context of Bluetooth.

Hence, because we are sending and receiving messages, there is an addressing scheme. There are different ways of addressing messages, as explained next.

Unicast addresses are present in every type of device and have a unique ID for that device. Surprisingly, these are hardly ever used, because most of the addressing of messages is done to sets of messages, even if a set contains only one device.

Groups of devices can be addressed in two ways using address types called group addresses and virtual addresses. Group addresses are those that are most commonly used.

Group addresses define some kind of set or collection of devices. More often than not, these are associated with physical parts of a building like a certain room or a certain type of device in that room, so all lights in the main